

ECE 484: Principles of Safe Autonomy
Lecture 10: Control (part 3)
Linear systems, Stability, Control design
by Eigenvalue placement

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Slides adapted in part from materials by Prof. Sayan Mitra.

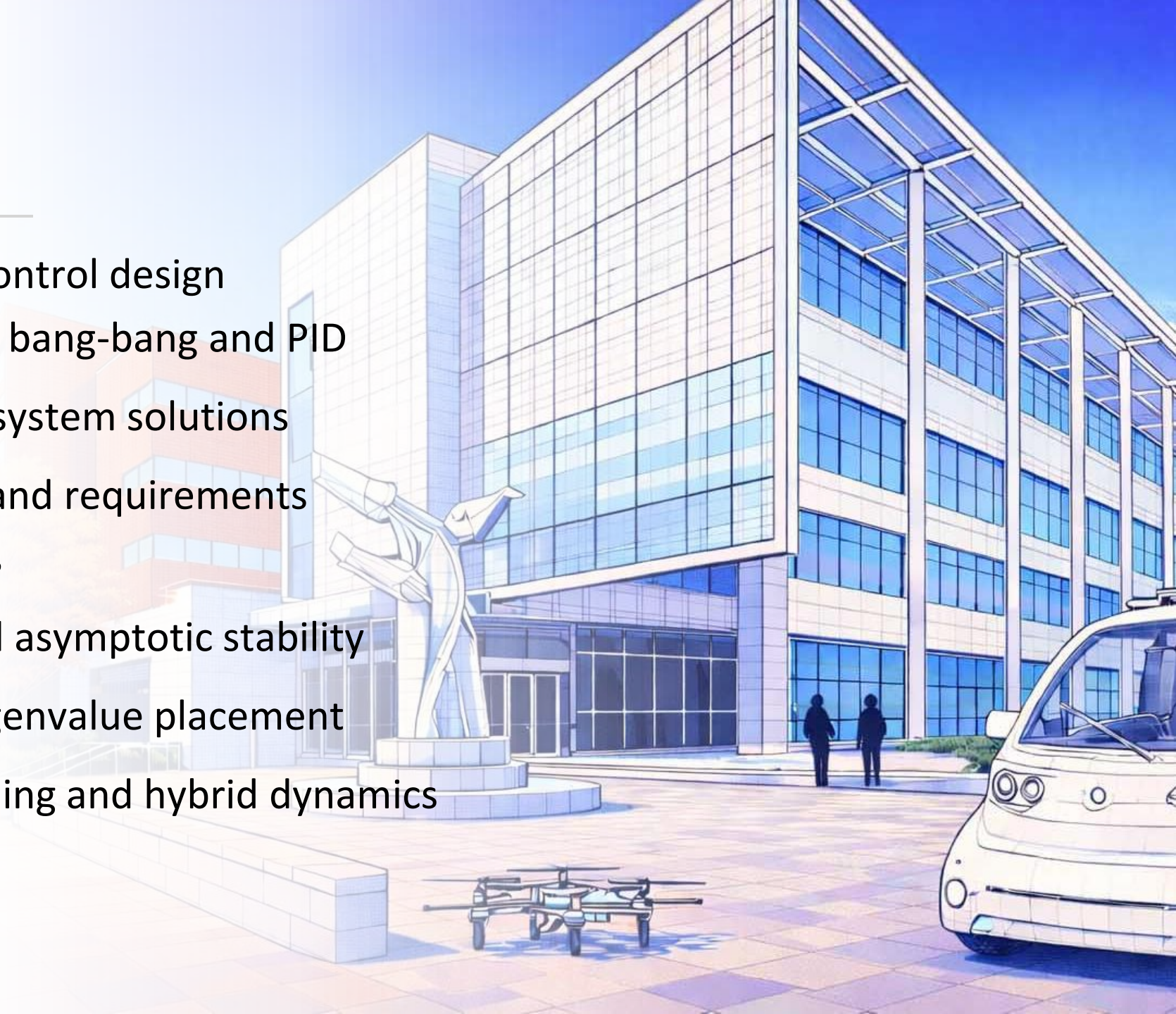
Outline

Last lecture: modeling and control design

- Differential equations; bang-bang and PID
- Path following; linear-system solutions

This lecture: linear systems and requirements

- Review: linear systems
- Equilibria; **stability** and asymptotic stability
- State feedback and eigenvalue placement
- Vehicle control; switching and hybrid dynamics



Linear dynamical system

A **linear model** is linear jointly in state and input:

$$\dot{x}(t) = f(t, x(t), u(t)) = A(t)x(t) + B(t)u(t)$$

$x(t) \in \mathbb{R}^n$: state; $u(t) \in \mathbb{R}^m$: input.

$A(t) \in \mathbb{R}^{n \times n}$, $B(t) \in \mathbb{R}^{n \times m}$: model matrices.

n : number of states; m : number of inputs; t_0 : initial time.

Given $x(t_0) = x_0$ and input signal u , write the solution as $\xi(t, t_0, x_0, u)$.

Superposition combines initial states AND inputs.

For one fixed nonzero input, the solution set is **affine**: a linear set shifted by one input response.

Quick check: which models are linear?

State and input can vary; **time-dependent coefficients are allowed**.

A. Damped spring

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -2x_1 - 0.4x_2 + u\end{aligned}$$

B. Coefficient changes with time

$$\dot{x} = -(1 + t)x + u$$

C. Pendulum-like restoring force

$$\dot{x} = -\sin x + u$$

D. Constant background force

$$\dot{x} = -x + u + 1$$

For each: linear or not linear?

Solutions of linear systems: continuity

$$\dot{x}(t) = A(t)x(t) + B(t)u(t)$$

Assume continuous A, B and **piecewise continuous** input u .

D_u : the input's jump times. $x(t_0) = x_0$; $\xi(t, t_0, x_0, u)$ is the solution.

1. Time continuity: the state stays continuous, even when the input jumps.

It is differentiable for every $t \notin D_u$.

2. Initial-state continuity: nearby initial states give nearby states at any fixed time.

Quick check: can a force step make position jump instantly?

Solutions of linear systems: superposition

For the same linear model, keep the same initial time t_0 .

3. Decomposition: initial conditions and applied input contribute separately.

$$\xi(t, t_0, x_0, u) = \xi(t, t_0, x_0, 0) + \xi(t, t_0, 0, u)$$

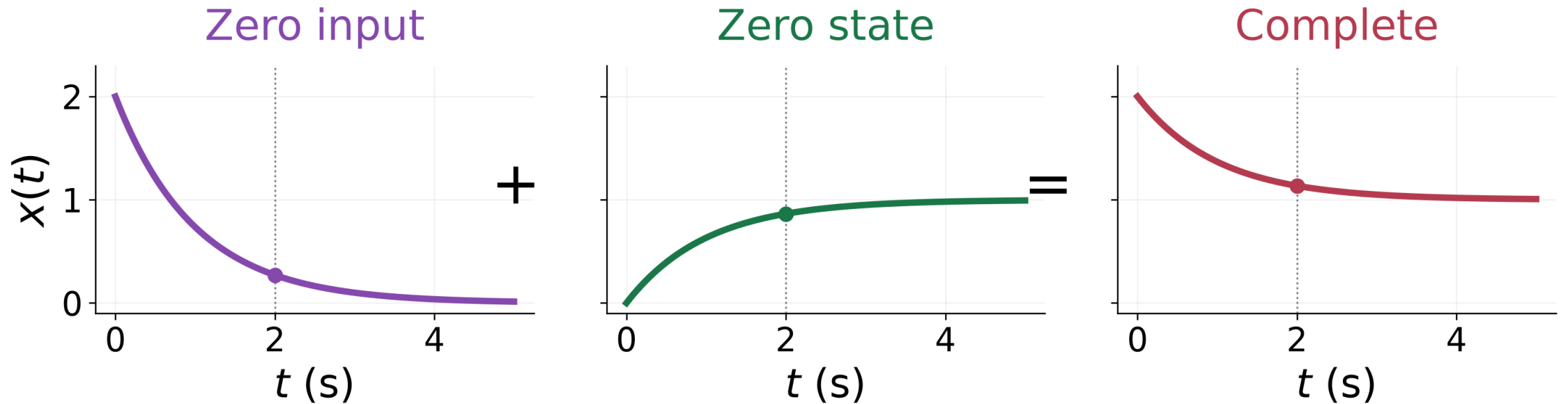
4. Superposition: combine both initial states and input signals.

$$\begin{aligned} &\xi(t, t_0, a_1 x_{01} + a_2 x_{02}, a_1 u_1 + a_2 u_2) \\ &= a_1 \xi(t, t_0, x_{01}, u_1) + a_2 \xi(t, t_0, x_{02}, u_2) \end{aligned}$$

$a_1, a_2 \in \mathbb{R}$; $x_{01}, x_{02} \in \mathbb{R}^n$; u_1, u_2 are m -component inputs.

Decomposition: two causes of motion add

Same model: $\dot{x} = -x + u$; start at $x(0) = 2$; apply $u(t) = 1$.



Zero input: $2e^{-t}$

Zero state: $1 - e^{-t}$

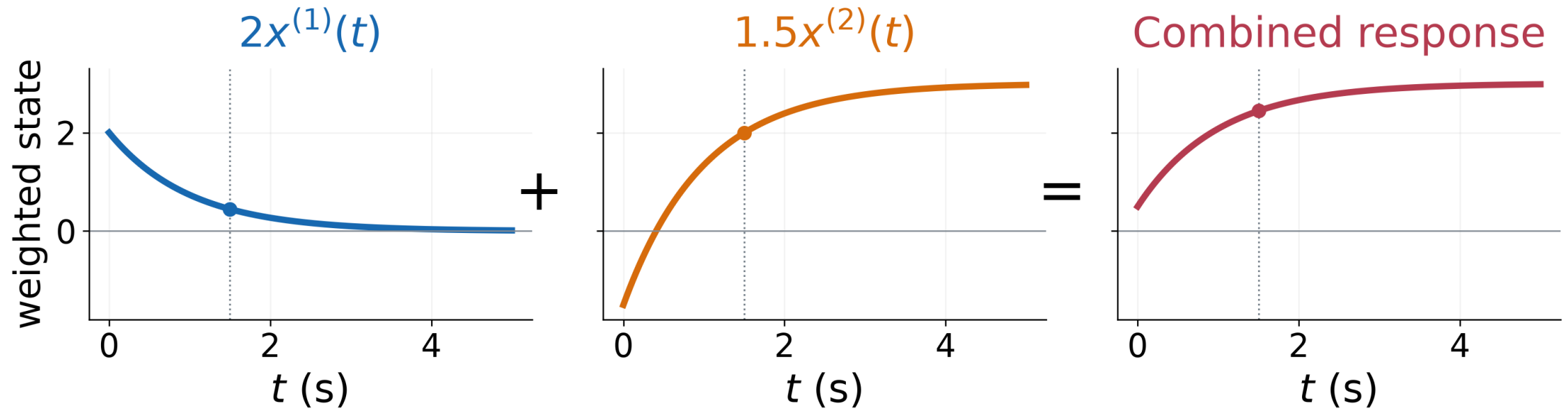
Complete: $1 + e^{-t}$

Superposition: scale, then add the responses

Same model: $\dot{x} = -x + u$. Superscripts (1), (2) label two experiments.

$$x_{01} = 1, u_1 = 0$$
$$x^{(1)}(t) = e^{-t}$$

$$x_{02} = -1, u_2 = 2$$
$$x^{(2)}(t) = 2 - 3e^{-t}$$



Combined: $x_0 = 2x_{01} + 1.5x_{02} = 0.5$, $u = 2u_1 + 1.5u_2 = 3$.

Example

x_1 : position, $x_1(0) = 0$ m; x_2 : velocity, $x_2(0) = -2$ m/s.
 $u(t) = 6$ N for $t \geq 0$; $m = 3$ kg, $b = 0.6$ N s/m, $k = 1.5$ N/m.

$$m\dot{x}_2 = u - bx_2 - kx_1$$
$$\dot{x}_1 = x_2$$

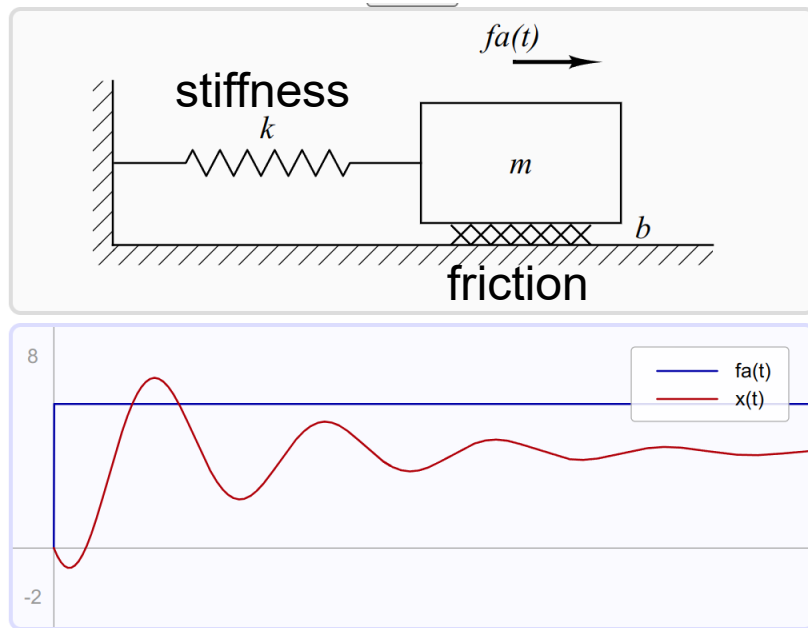
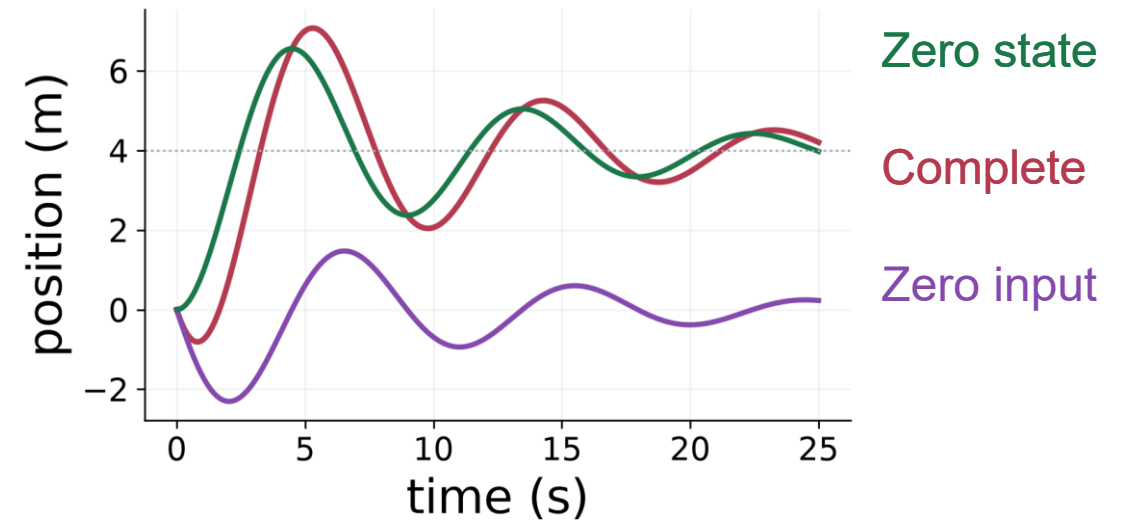


Figure labels: $fa(t) = u(t)$; $x(t) = x_1(t)$.

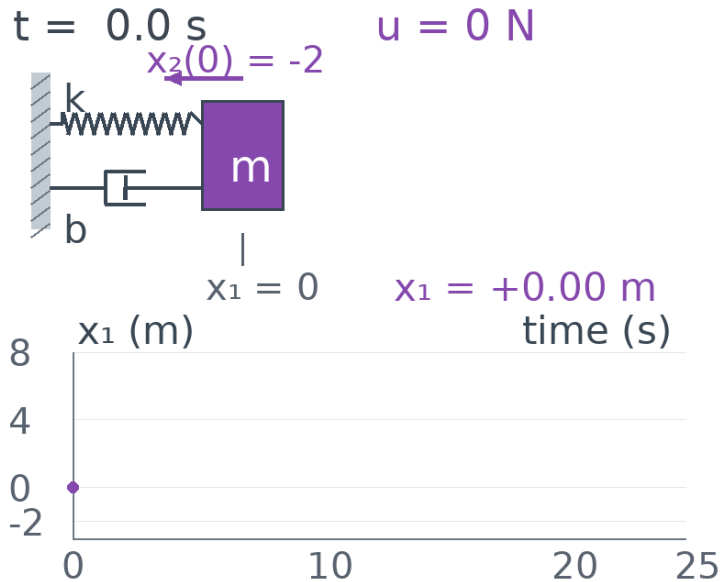


Spring and mass: watch the three responses

As before: $m = 3$ kg, $b = 0.6$ N s/m, $k = 1.5$ N/m; x_1 : position, x_2 : velocity.

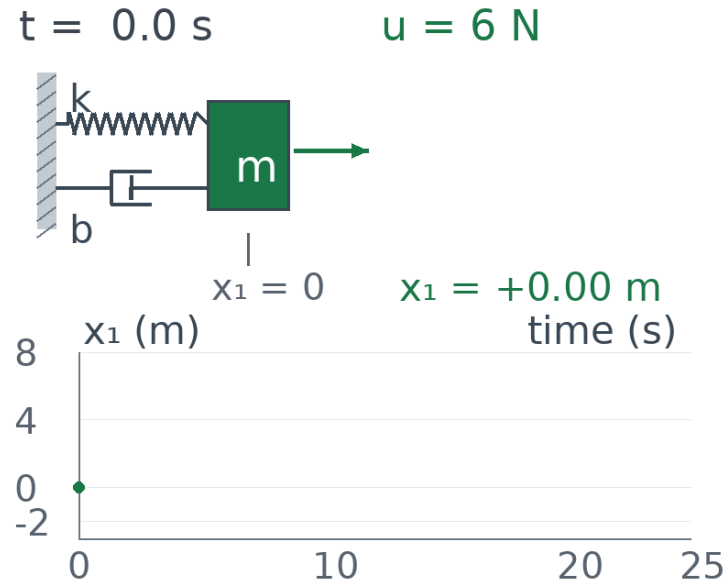
Zero input

$$x(0) = [0, -2]^T$$
$$u(t) = 0$$



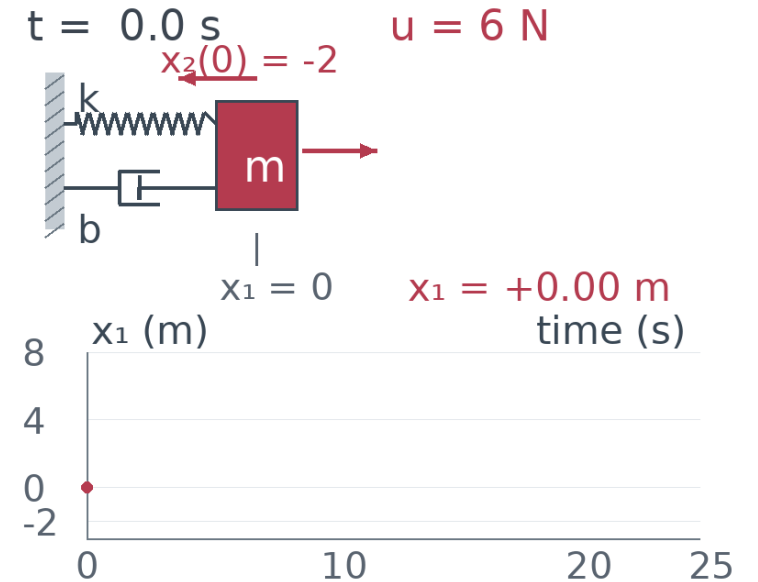
Zero state

$$x(0) = [0, 0]^T$$
$$u(t) = 6$$
 N



Complete

$$x(0) = [0, -2]^T$$
$$u(t) = 6$$
 N



Same clock and scale. At each time: zero input + zero state = complete.

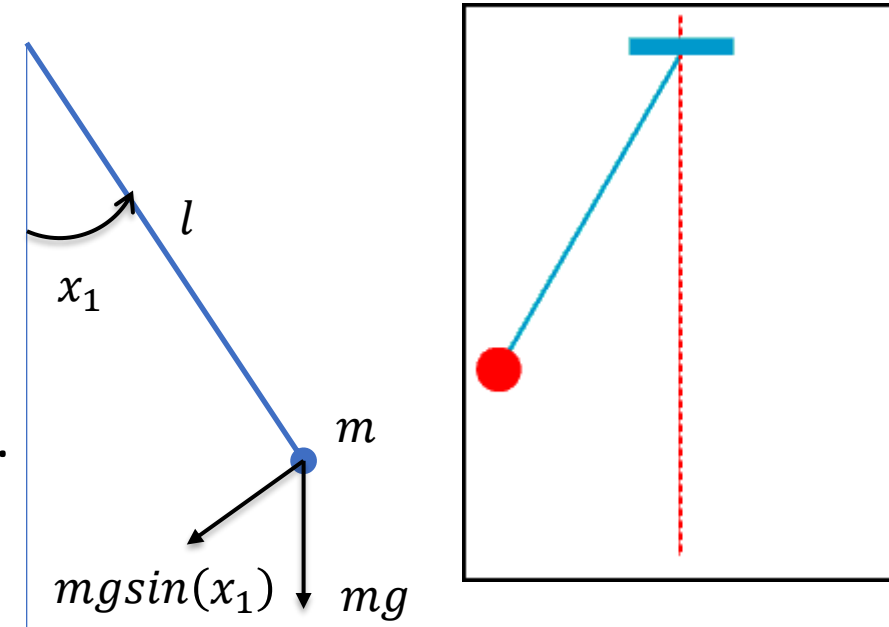
Revisit: pendulum dynamics

$x = [x_1, x_2]^T$: x_1 is angle; $x_2 = \dot{x}_1$ is angular velocity.

Autonomous: no external control input.

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{g}{l} \sin x_1 - \frac{k}{m} x_2 \\ \dot{x} &= \begin{bmatrix} x_2 \\ -\frac{g}{l} \sin x_1 - \frac{k}{m} x_2 \end{bmatrix}\end{aligned}$$

g : gravity; l : length; m : mass; k : tangential damping coefficient.



Not a linear system

Linear dynamical system and solutions

Linearize near an equilibrium, not over the whole angle range.

$A = \left. \frac{\partial f}{\partial x} \right|_{x_e}$: Jacobian (matrix of partial derivatives).

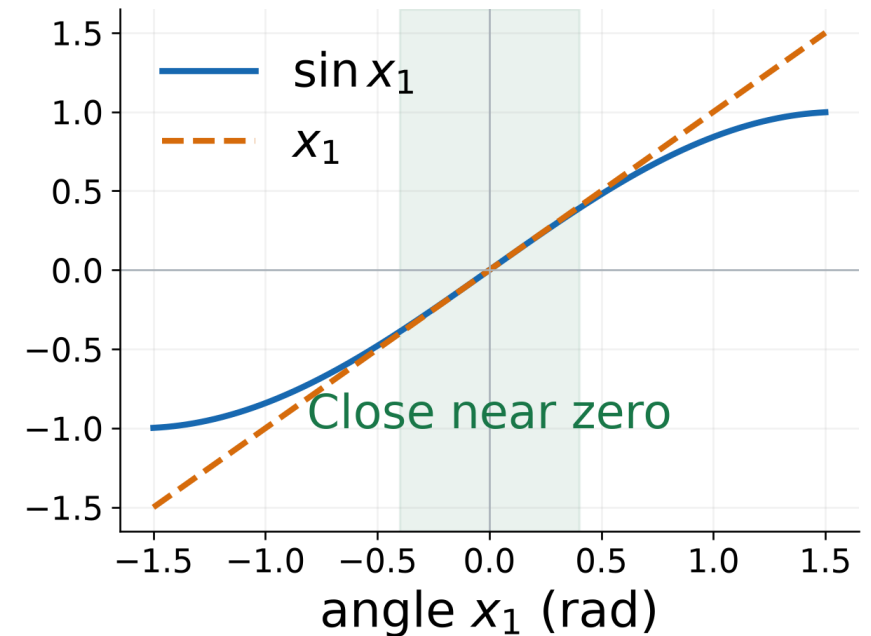
For the pendulum, $x_e = (0,0)$; locally $\sin x_1 \approx x_1$.

$$\dot{x} = \begin{bmatrix} x_2 \\ -\frac{g}{l} \sin x_1 - \frac{k}{m} x_2 \end{bmatrix}$$

$$\dot{x} \approx \begin{bmatrix} x_2 \\ -\frac{g}{l} x_1 - \frac{k}{m} x_2 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ -g/l & -k/m \end{bmatrix}$$

x is already a deviation at $x_e = 0$.



Solutions of Linear time invariant system

LTI: A and B are constant matrices.

$$\dot{x}(t) = Ax(t) + Bu(t); x(t_0) = x_0.$$

$$\xi(t, t_0, x_0, u) = e^{A(t-t_0)}x_0 + \int_{t_0}^t e^{A(t-\tau)}Bu(\tau) d\tau$$

No input: $\xi(t, t_0, x_0, 0) = e^{A(t-t_0)}x_0.$

$$e^{At} = I + At + \frac{(At)^2}{2!} + \dots = \sum_{k=0}^{\infty} \frac{(At)^k}{k!}$$

I : identity matrix; τ : integration variable; k : nonnegative integer.

The two terms are **zero input** and **zero state**, as in the spring example.

Example: Simple linear model of an economy

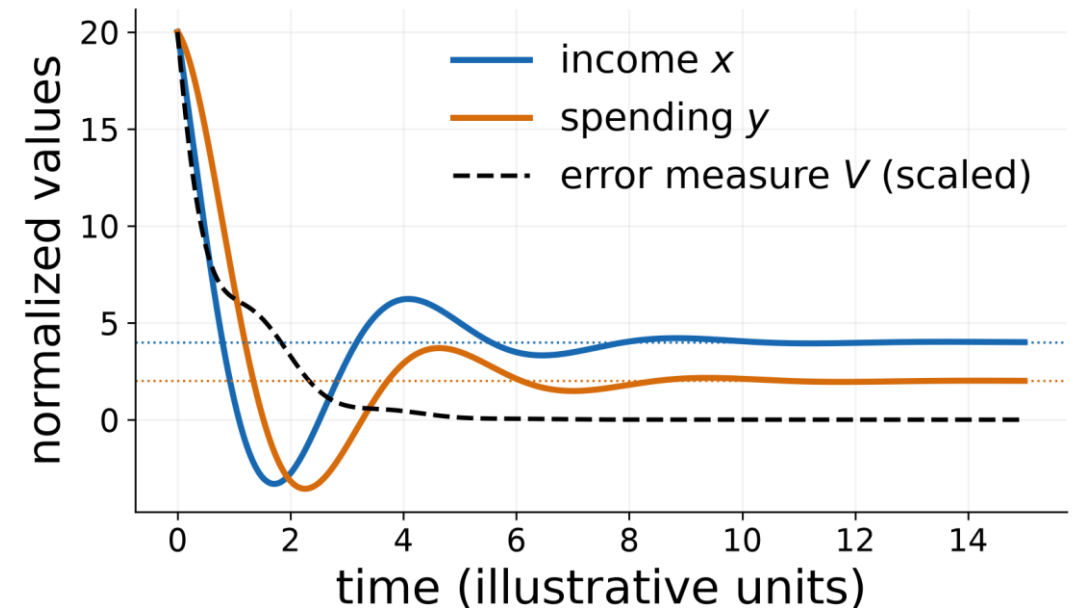
x : national income; y : consumer-spending rate.

g : government-expenditure input; α, β : model coefficients.

$$\begin{aligned}\dot{x} &= x - \alpha y \\ \dot{y} &= \beta(x - y - g) \\ \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} &= \begin{bmatrix} 1 & -\alpha \\ \beta & -\beta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} 0 \\ \beta \end{bmatrix} g\end{aligned}$$

Toy example: $\alpha = \beta = 2$, $g = 2$ gives **equilibrium** (4,2).

Dashed **V**: a weighted squared distance to equilibrium (scaled in the plot).



$$\dot{x}(t) = Ax(t) + Bu(t)$$

Example 2D LTI systems (assuming no control u)

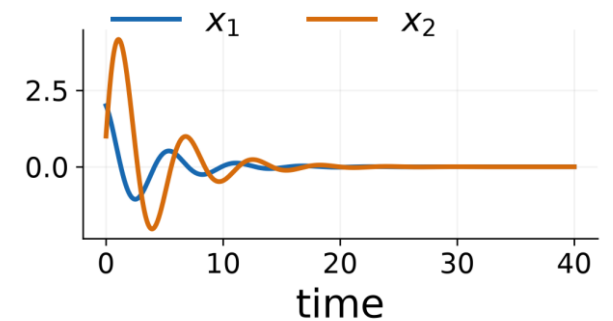
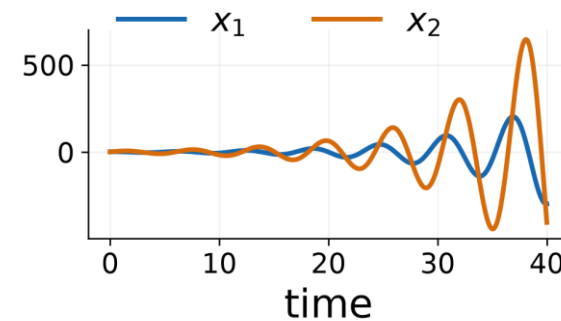
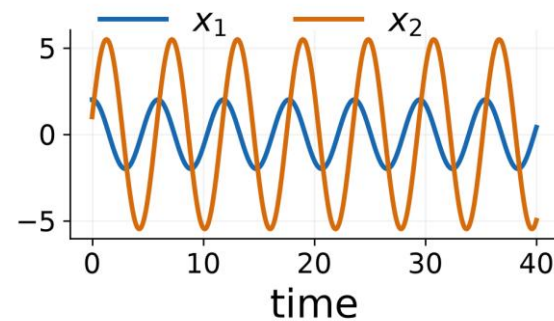
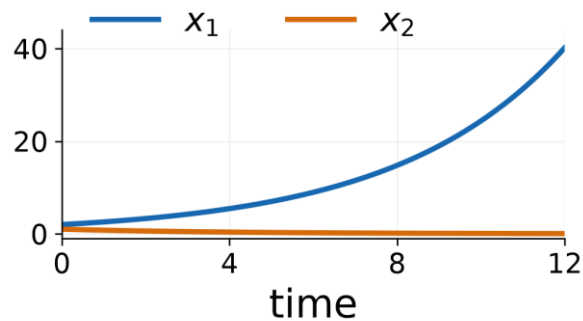
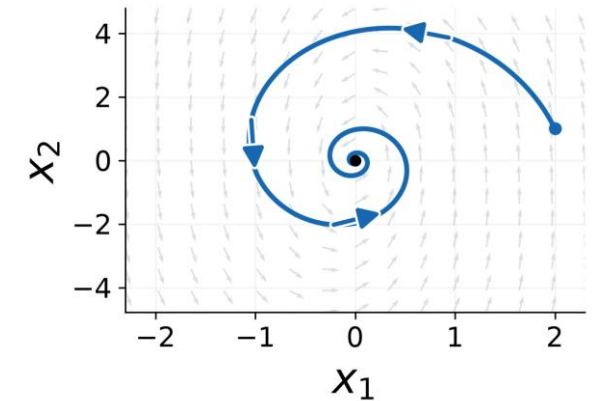
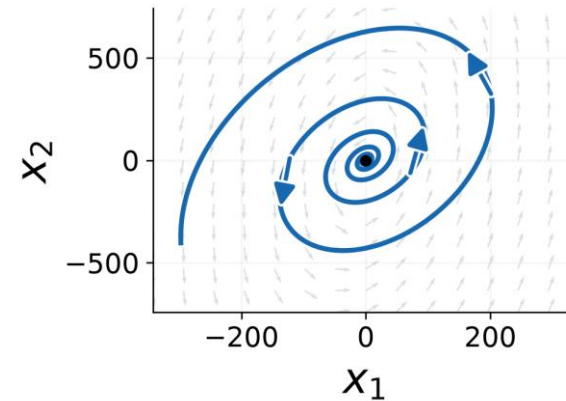
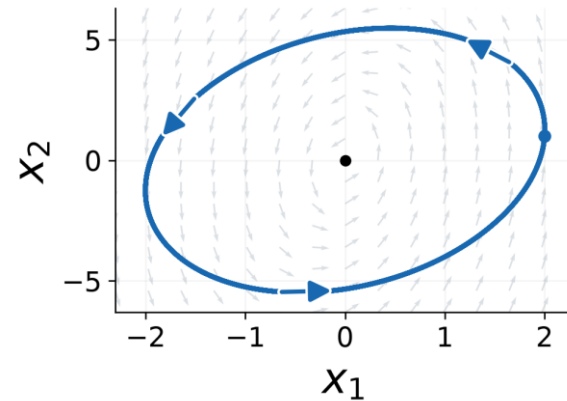
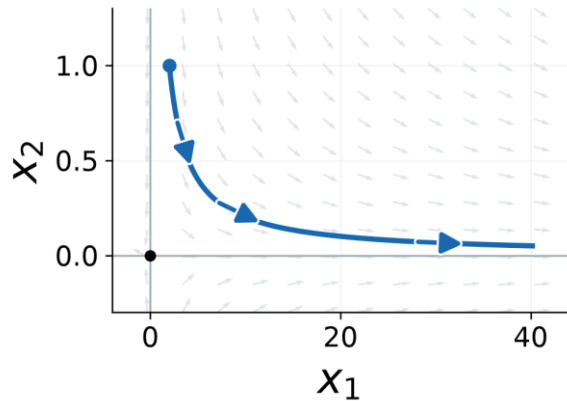
Try it yourself! <https://colab.research.google.com/drive/1TXjVYl8fHWWhhN72tPCEikMq5xXB2-QMI?usp=sharing>

$$A = \begin{bmatrix} 1/4 & 0 \\ 0 & -1/4 \end{bmatrix}$$

$$A = \begin{bmatrix} 1/4 & -2/5 \\ 3 & -1/4 \end{bmatrix}$$

$$A = \begin{bmatrix} 1/2 & -2/5 \\ 3 & -1/4 \end{bmatrix}$$

$$A = \begin{bmatrix} -1/4 & -2/5 \\ 3 & -1/4 \end{bmatrix}$$



Initial state: $x_1(0)=2$, $x_2(0)=1$; no input. Phase portrait above; time response below.

Requirements: Equilibria and Stability

Consider an **autonomous** system (no external input):

$\dot{x} = f(x)$; for a linear system, $f(x) = Ax$.

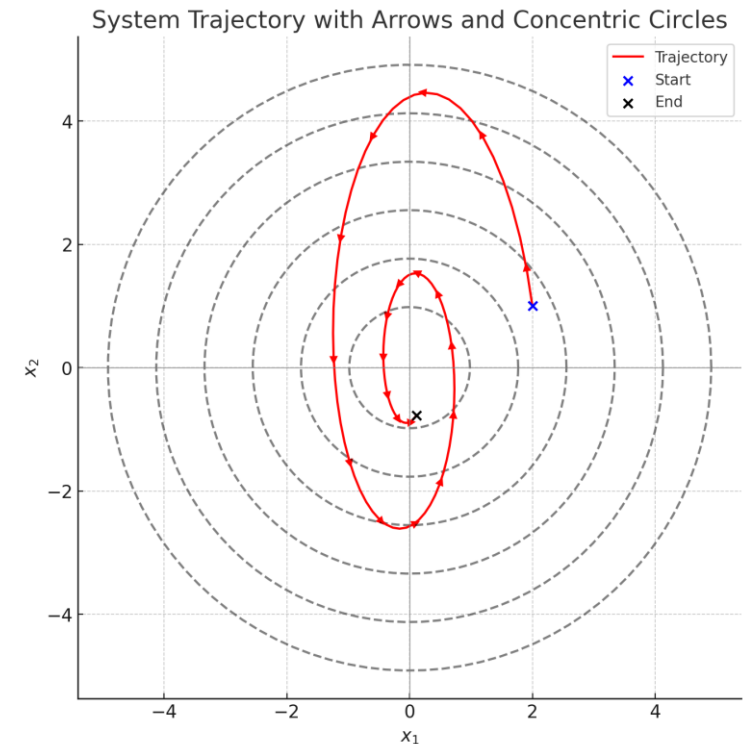
x^* is an **equilibrium** if $f(x^*) = 0$.

Shift coordinates: $z = x - x^*$, so the chosen equilibrium becomes $z = 0$.

Write $x(t) = \xi(t, 0, x_0, 0)$ after that shift, with $x(0) = x_0$.

$\|x(t)\|$: Euclidean distance to the origin (equilibrium).

Question: does a nearby start **stay** nearby, **return**, or move **away**?



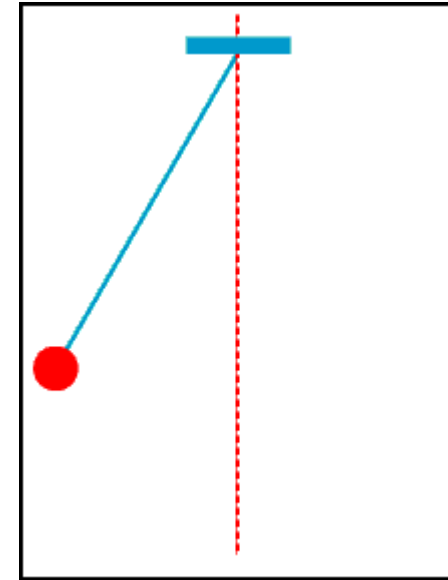
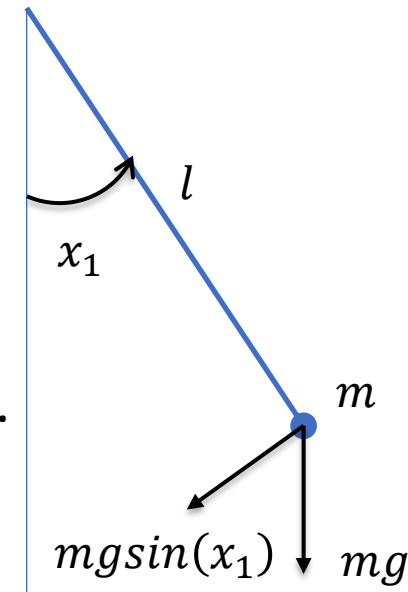
Revisit: pendulum equilibria

$x = [x_1, x_2]^T$: x_1 is angle; $x_2 = \dot{x}_1$ is angular velocity.

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g : gravity; l : length; m : mass; k : tangential damping coefficient.



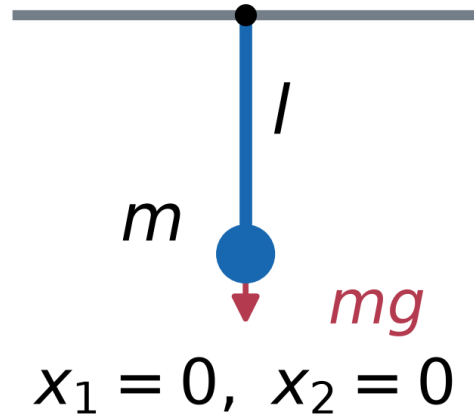
Which states are equilibria? Explain using both derivatives.

A: $(0,0)$ B: $(\pi, 0)$ C: $(\pi/2, 0)$ D: $(0,1)$

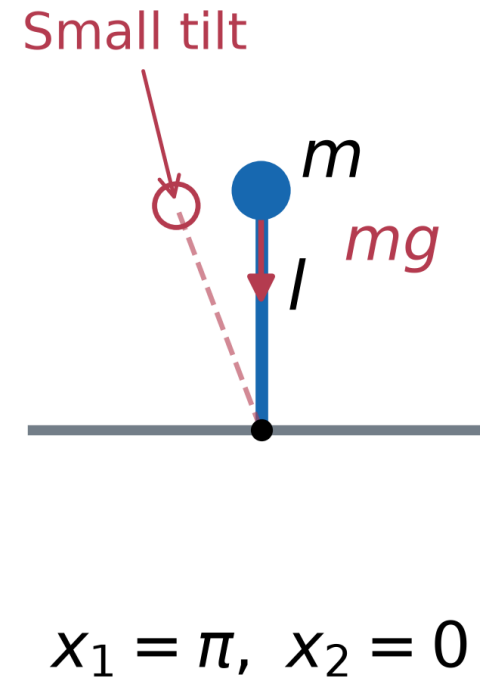
Pendulum equilibria: down and upright

Both vertical positions can remain still when the angular velocity is zero.

Down

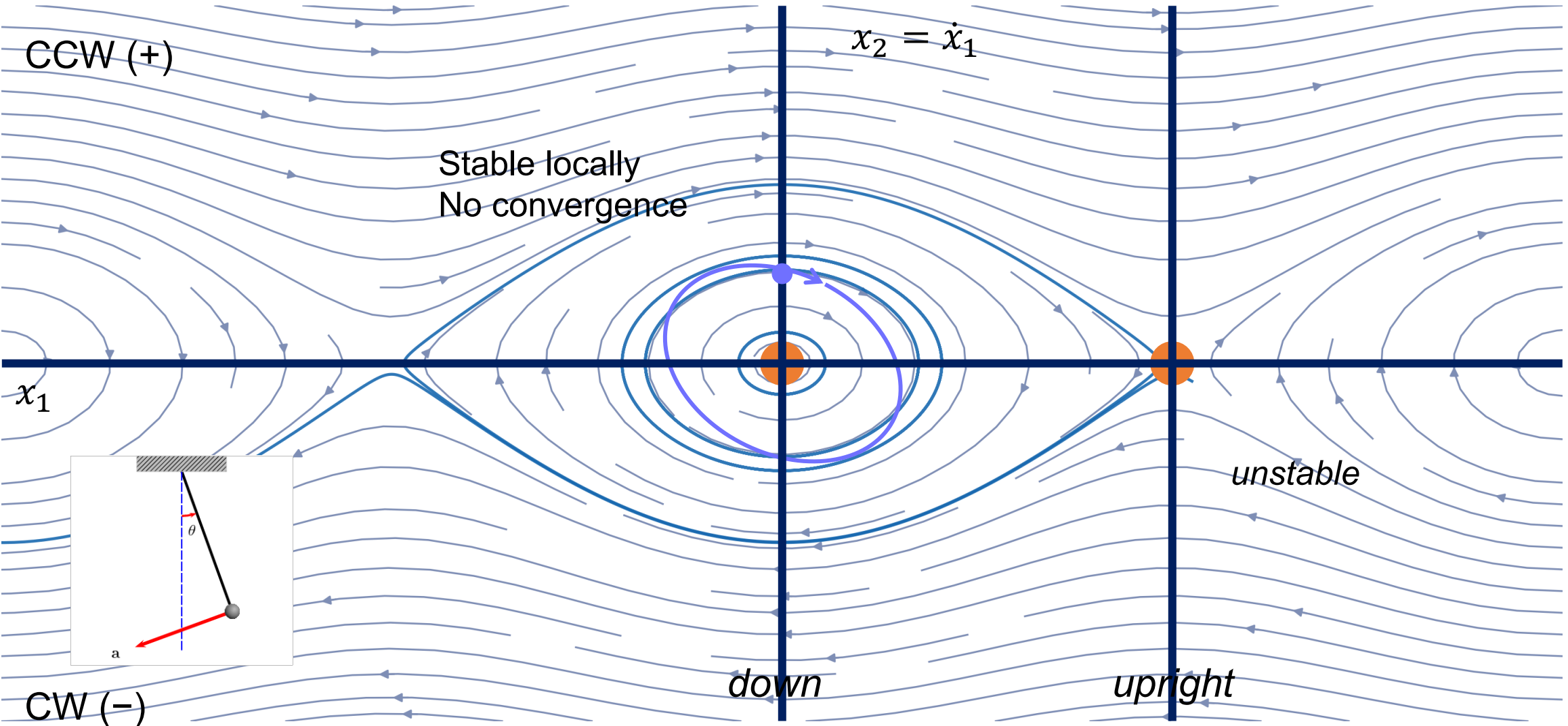


Upright



Equilibrium: can stay still. **Stability:** small disturbances stay small.

Pendulum without friction: staying nearby



Lyapunov stability

Lyapunov stability: sufficiently small initial errors stay small **for all future time**.

For every tolerance $\varepsilon > 0$, choose a starting radius $\delta_\varepsilon > 0$ such that

$$\|x_0\| \leq \delta_\varepsilon \Rightarrow \|x(t)\| \leq \varepsilon \quad \text{for all } t \geq 0.$$

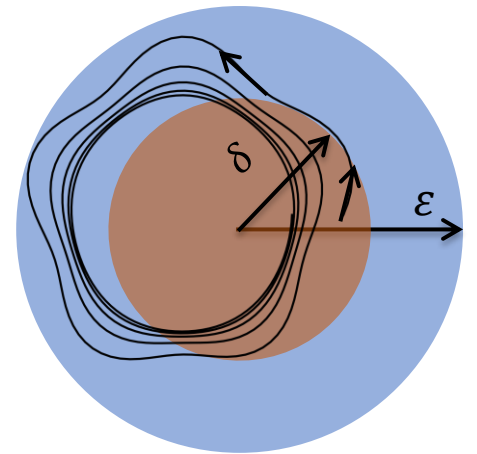
$x(0) = x_0$; $\|\cdot\|$: Euclidean norm. The choice of δ_ε may depend on ε .

Exercise: how does this relate to reachable states?

Every state reachable from the inner starting ball lies in the outer tolerance ball.



A. M. Lyapunov



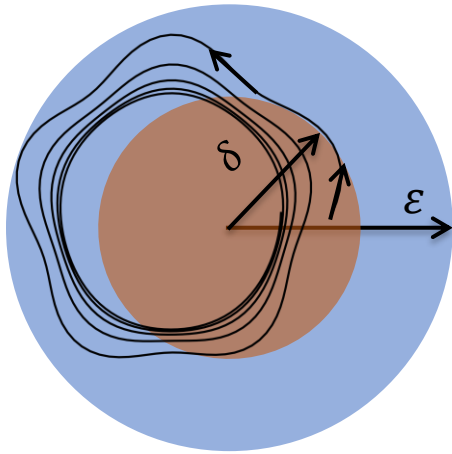
Asymptotic stability

Asymptotic stability = stay nearby AND return.

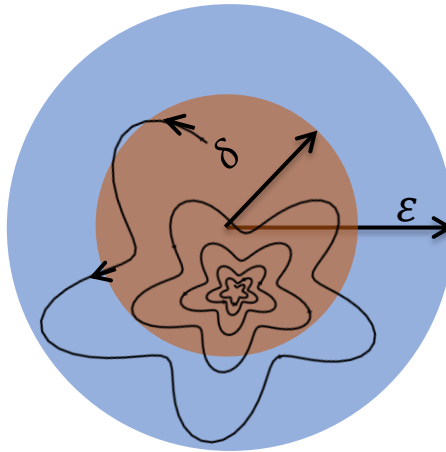
Require Lyapunov stability and one attraction radius $\delta_a > 0$ such that **every** start satisfies

$$\|x_0\| < \delta_a \Rightarrow \lim_{t \rightarrow \infty} \|x(t)\| = 0.$$

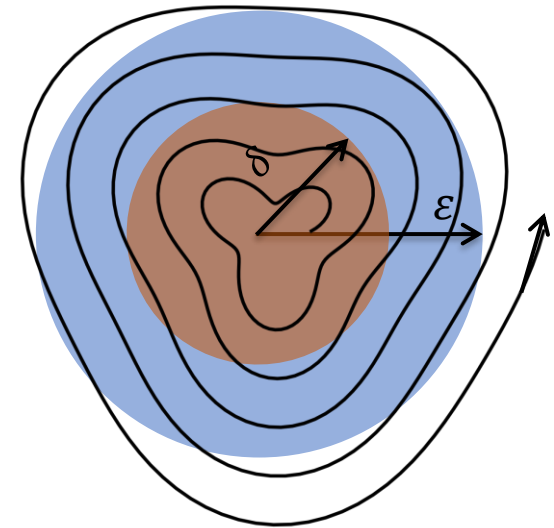
Global: the convergence condition holds for every initial state.



Lyapunov stable



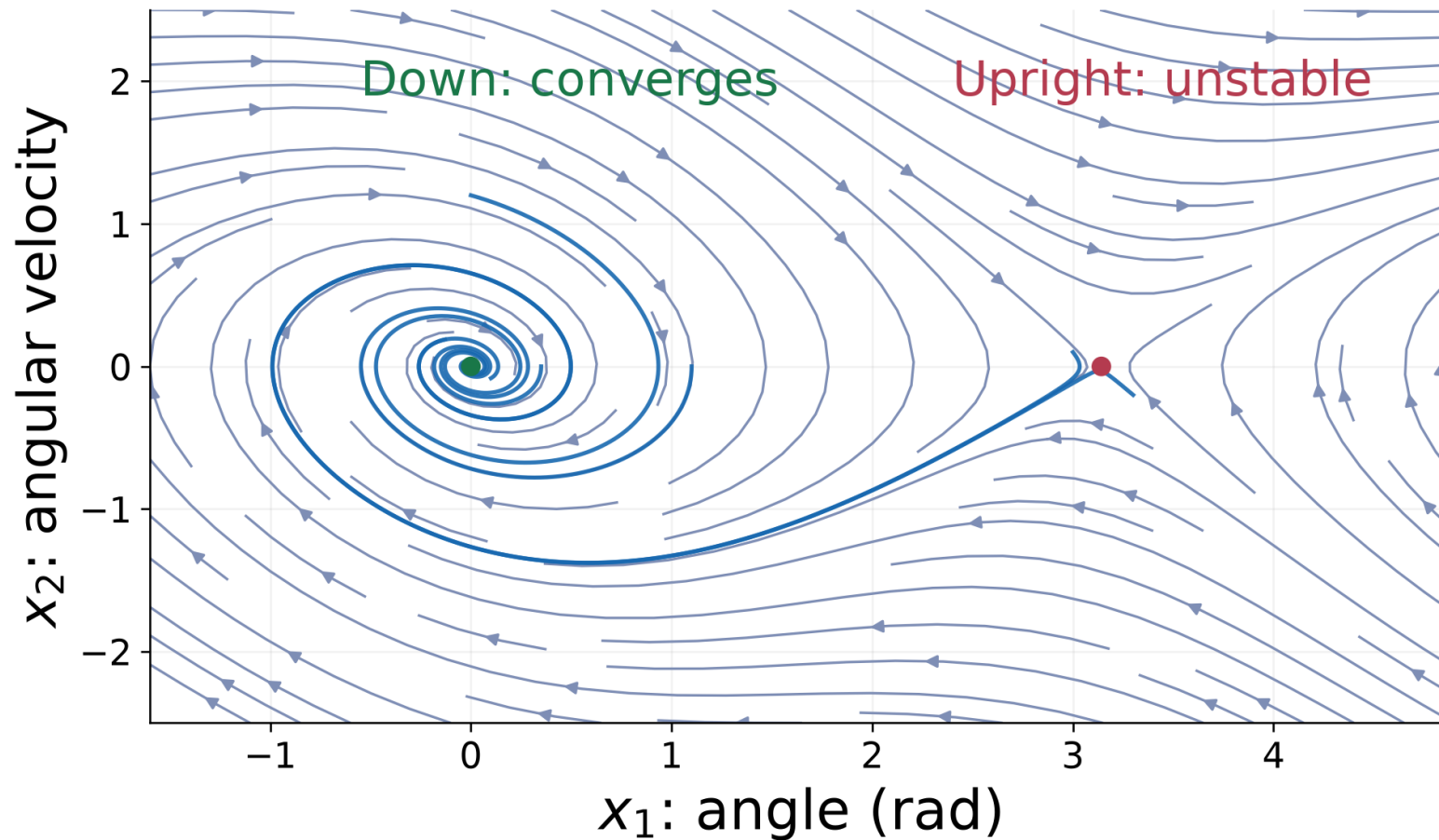
Asymptotically stable



Unstable

Phase portrait of pendulum with friction

For each orientation: stable, asymptotically stable, or unstable?



Answer: down is locally asymptotically stable; upright is unstable.

Butterfly

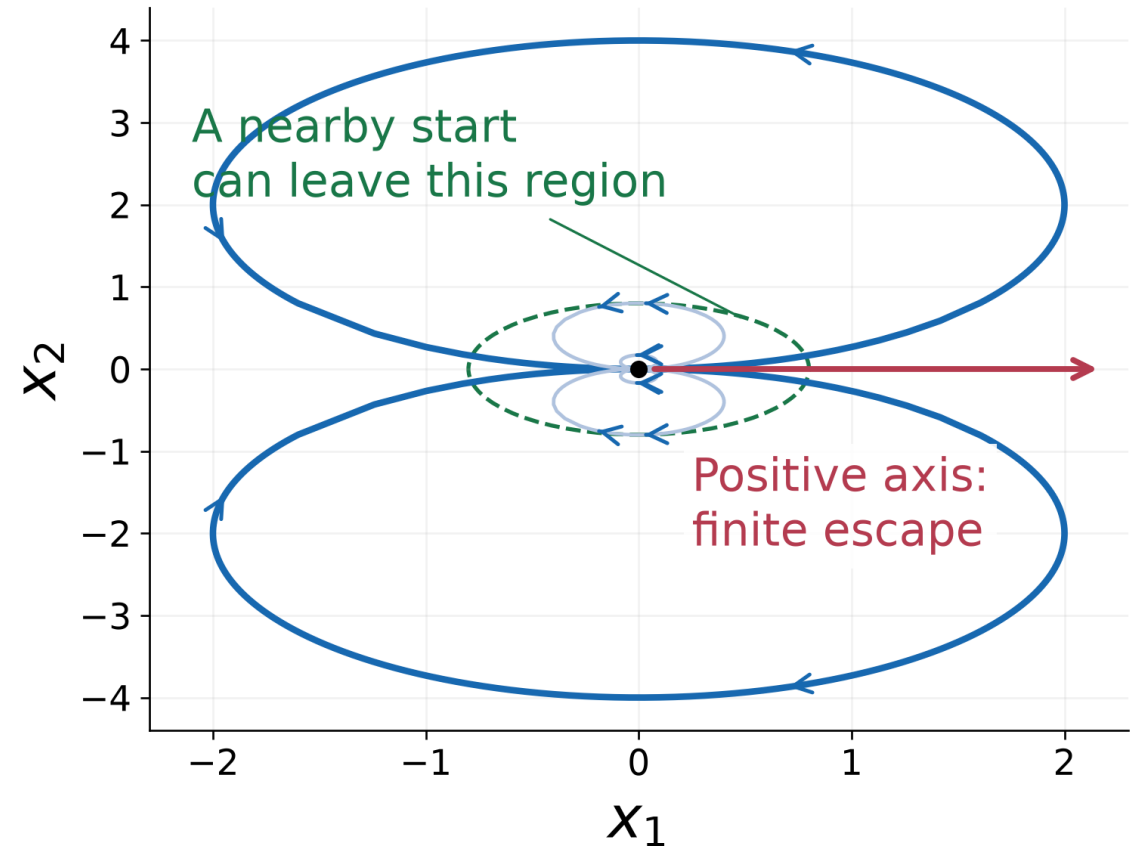
$$\begin{aligned}\dot{x}_1 &= x_1^2 - x_2^2 \\ \dot{x}_2 &= 2x_1x_2\end{aligned}$$

Many nearby trajectories return to the origin.

Some first make a **large excursion**: the origin is not Lyapunov stable.

On the positive x_1 -axis, the state escapes in finite time.

A returning trajectory is not a stability proof.



Verifying Stability for Linear Systems

For $\dot{x} = Ax$, the origin is an equilibrium.

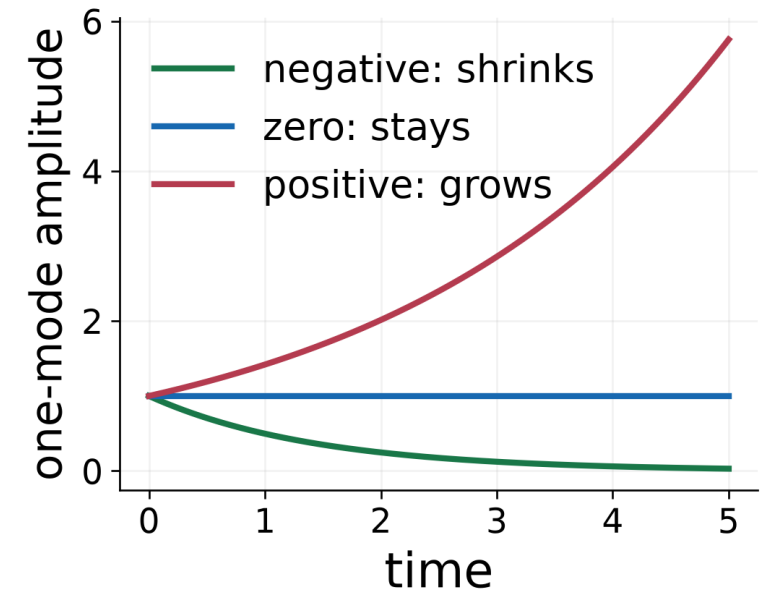
eigenvalue λ : solve $\det(\lambda I - A) = 0$.

Asymptotically stable iff every $\operatorname{Re}\lambda < 0$. This is called **Hurwitz**.

Lyapunov stable iff every $\operatorname{Re}\lambda \leq 0$ AND the zero-real-part Jordan blocks have size 1.

$\operatorname{Re}\lambda$: real part; $\det$: determinant; $i^2 = -1$.

Zero real parts need care: a small velocity can cause drift (next page).



Why zero eigenvalues need care*

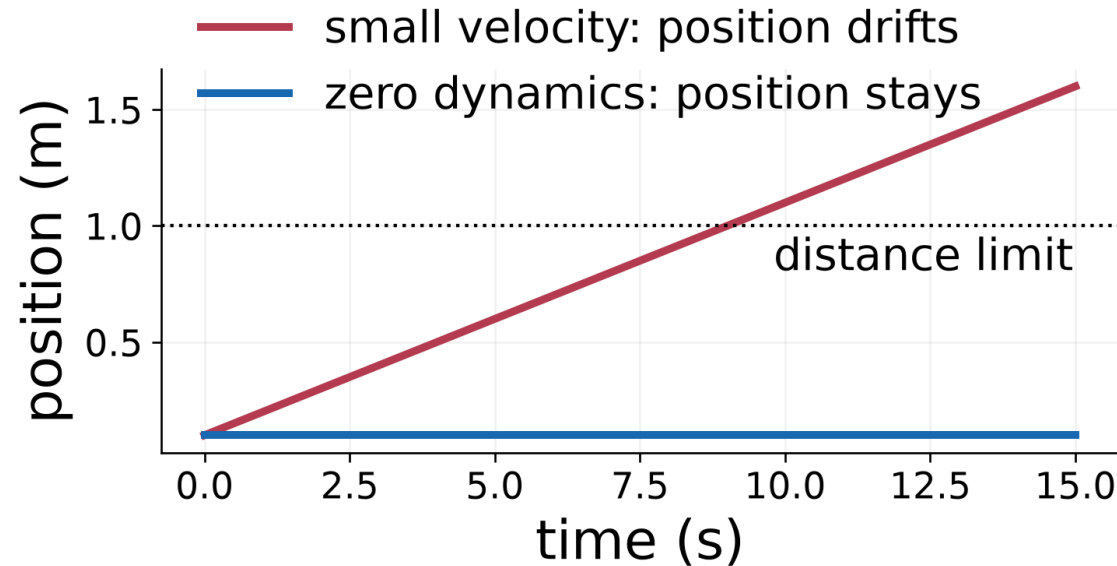
Two systems have eigenvalues 0,0:

Frozen: $\dot{x}_1 = 0, \dot{x}_2 = 0$ — every nearby start stays nearby.

Drifting: $\dot{x}_1 = x_2, \dot{x}_2 = 0$ — a tiny velocity keeps moving position.

$$x_1(t) = x_1(0) + tx_2(0), x_2(t) = x_2(0).$$

Question: can choosing an even smaller initial velocity prevent eventual drift?



* Not in your exam

Jordan decomposition: the two previous examples*

Change coordinates: $z = Px$ gives $\dot{z} = Jz$, where $J = PAP^{-1}$.

P : invertible coordinate change. For these two examples, $P = I$, so $J = A$.

Frozen: $\dot{x}_1 = 0, \dot{x}_2 = 0$

$$J = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Two blocks, each 1×1 .

$$x_1(t) = x_1(0); x_2(t) = x_2(0).$$

Drifting: $\dot{x}_1 = x_2, \dot{x}_2 = 0$

$$J = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

One block, 2×2 .

$$x_1(t) = x_1(0) + tx_2(0).$$

Same eigenvalues: **0 and 0**. Different coupling; different stability.

The off-diagonal **1** means: constant velocity keeps changing position.

* Not in your exam

Jordan decomposition: general form*

Using complex numbers when needed, an invertible P gives $PAP^{-1} = J$.

$$PAP^{-1} = J = \begin{bmatrix} J_1 & 0 & \cdots & 0 \\ 0 & J_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & J_r \end{bmatrix}$$

$$J_i = \begin{bmatrix} \lambda_i & 1 & 0 & \cdots \\ 0 & \lambda_i & 1 & \cdots \\ \vdots & \vdots & \ddots & 1 \\ 0 & 0 & \cdots & \lambda_i \end{bmatrix}$$

J_i : one Jordan block; λ_i : its eigenvalue; r : number of blocks.

The off-diagonal 1s couple modes and can create powers of time.

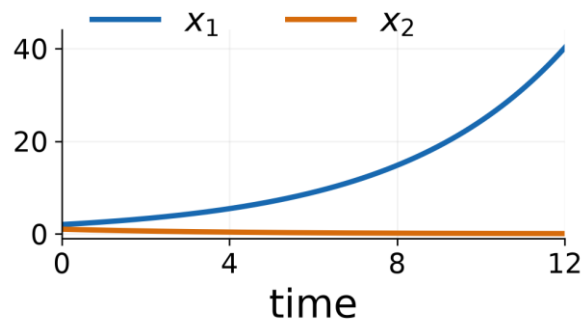
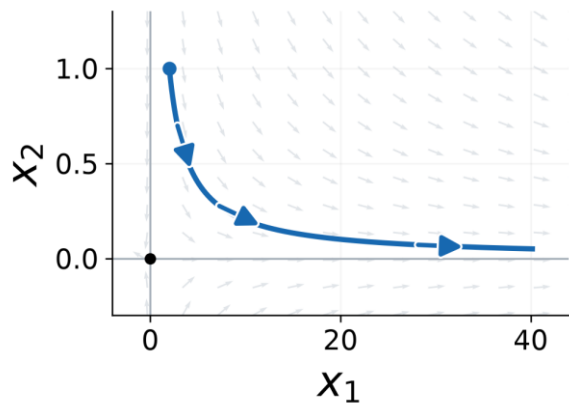
At zero real part, a block larger than 1 can produce drift; see the preceding example.

* Not in your exam

The same four systems: what their eigenvalues predict

$$\dot{x} = Ax, A = \begin{bmatrix} 1/4 & 0 \\ 0 & -1/4 \end{bmatrix}$$

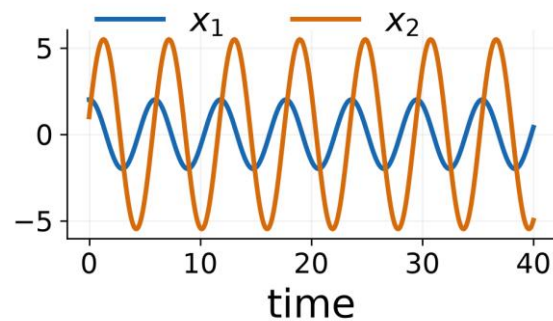
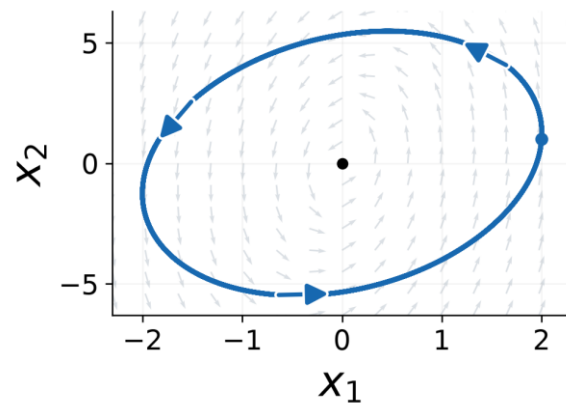
$$\lambda_1 = 1/4, \lambda_2 = -1/4$$



$$\dot{x} = Ax, A = \begin{bmatrix} 1/4 & -2/5 \\ 3 & -1/4 \end{bmatrix}$$

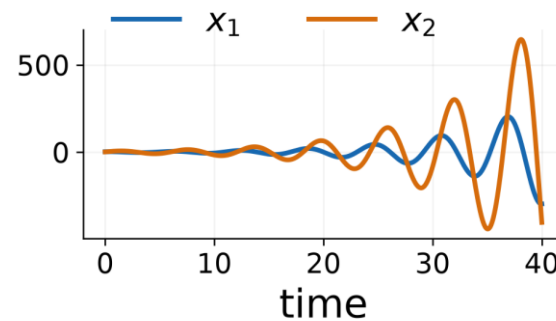
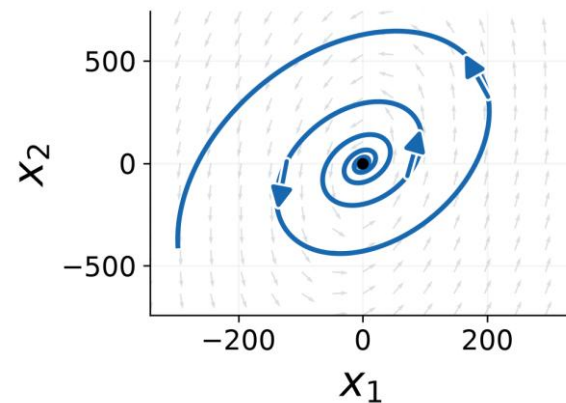
$$\lambda_{1,2} = \pm 1.066i$$

Zero-real-part blocks: size 1.



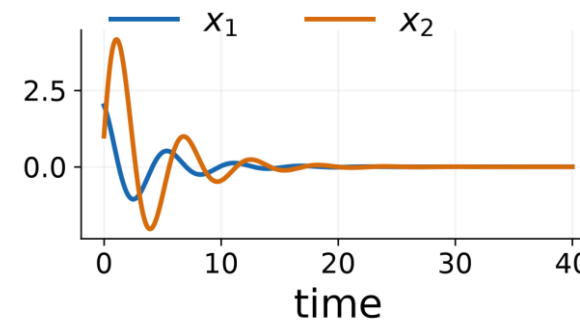
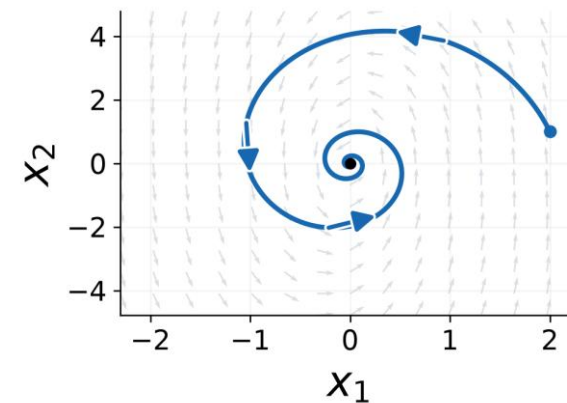
$$\dot{x} = Ax, A = \begin{bmatrix} 1/2 & -2/5 \\ 3 & -1/4 \end{bmatrix}$$

$$\lambda_{1,2} = 0.125 \pm 1.029i$$



$$\dot{x} = Ax, A = \begin{bmatrix} -1/4 & -2/5 \\ 3 & -1/4 \end{bmatrix}$$

$$\lambda_{1,2} = -0.250 \pm 1.095i$$



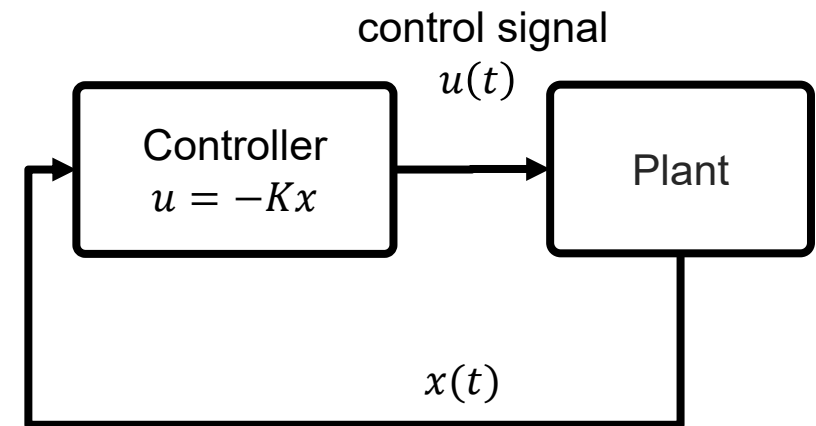
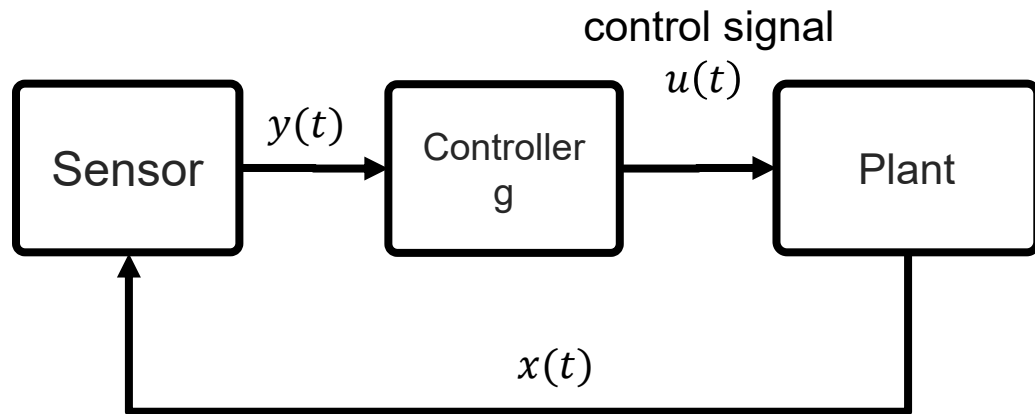
Initial state: $x_1(0)=2, x_2(0)=1$; no input. Phase portrait above; time response below.

Control design by Eigenvalue placement

Plant: $\dot{x} = Ax + Bu, x \in \mathbb{R}^n, u \in \mathbb{R}^m$.

Goal: stabilize the equilibrium at the origin using **full-state feedback**.

Choose $u = g(x) = -Kx$, where $K \in \mathbb{R}^{m \times n}$.



Full-state sensing: the measurement y equals x (or all state components are estimated).

Control design by Eigenvalue placement

Plant: $\dot{x} = Ax + Bu$, $x \in \mathbb{R}^n$, $u \in \mathbb{R}^m$.

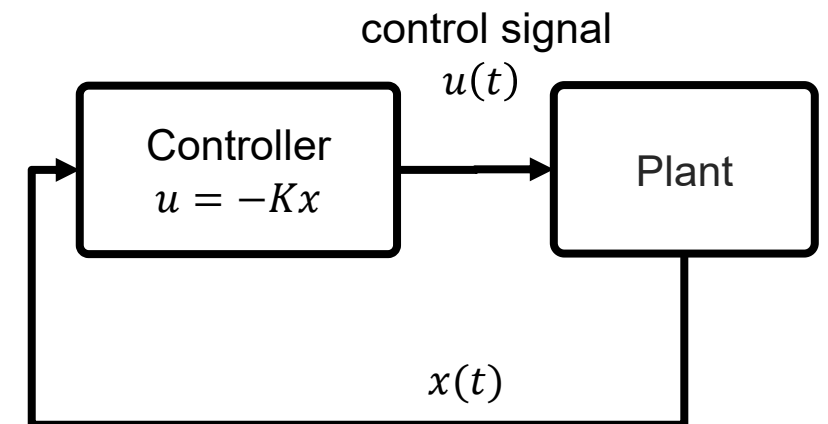
Goal: stabilize the equilibrium at the origin using **full-state feedback**.

Choose $u = g(x) = -Kx$, where $K \in \mathbb{R}^{m \times n}$.

$$\dot{x} = Ax - BKx = (A - BK)x = A_{cl}x$$

$A_{cl} := A - BK$: closed-loop state matrix.

Choose feasible K so every $\text{Re}\lambda(A_{cl}) < 0$.



Control design by Eigenvalue placement

$$A = \begin{bmatrix} 0 & v \\ 1 & 1/2 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}; v > 1/3 \text{ is a model parameter.}$$

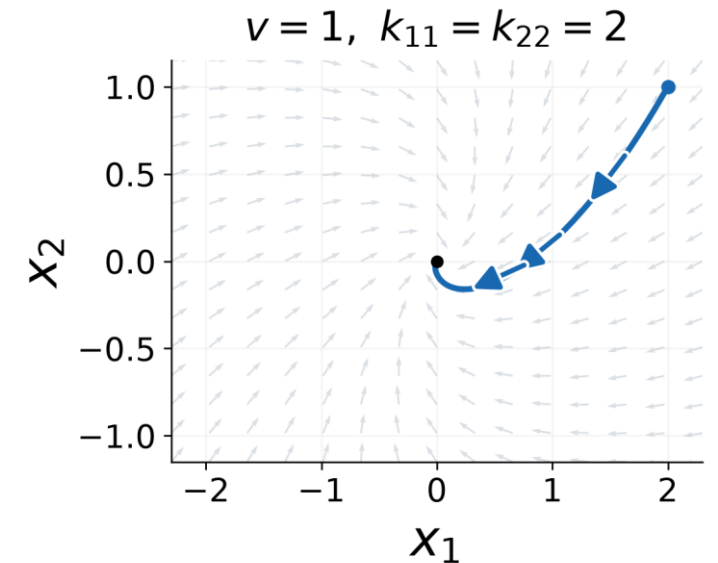
$$u = -Kx, K = \begin{bmatrix} k_{11} & 0 \\ 0 & k_{22} \end{bmatrix} \text{ (a diagonal design choice).}$$

$$BK = \begin{bmatrix} k_{11} & 0 \\ k_{11} & k_{22} \end{bmatrix}$$

$$\dot{x} = A_{cl}x, A_{cl} = A - BK = \begin{bmatrix} -k_{11} & v \\ 1 - k_{11} & 1/2 - k_{22} \end{bmatrix}$$

$$\text{Example: } v = 1, k_{11} = k_{22} = 2 \text{ gives } A_{cl} = \begin{bmatrix} -2 & 1 \\ -1 & -1.5 \end{bmatrix}.$$

Next: which gains make its eigenvalues stable?



Eigenvalue placement: choose the gains

Find roots of $\det(\lambda I - A_{cl}) = 0$:

2×2 determinant rule: diagonal product minus off-diagonal product.

$$\det \begin{bmatrix} \lambda + k_{11} & -v \\ -1 + k_{11} & \lambda - 1/2 + k_{22} \end{bmatrix} = 0$$

Rearrange as $\lambda^2 + b\lambda + c = 0$

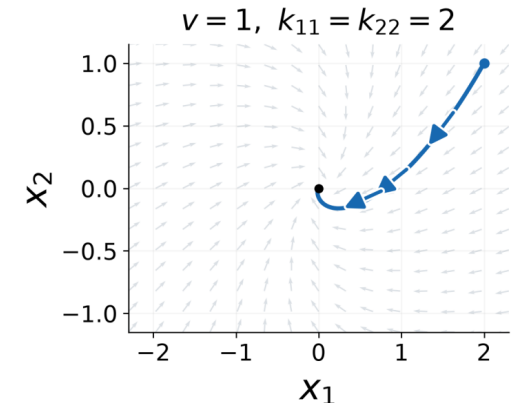
where $b = k_{11} + k_{22} - 1/2$

$c = k_{11}k_{22} - k_{11}/2 - v + vk_{11}$

To ensure **negative real parts**: $b > 0$ AND $c > 0$.

Root sum = $-b < 0$; root product = $c > 0$.

With $k_{11} = k_{22} = 2$: $b = 3.5$, $c = 3 + v > 0$ for $v > 1/3$.



Can the actuator influence the unstable motion?

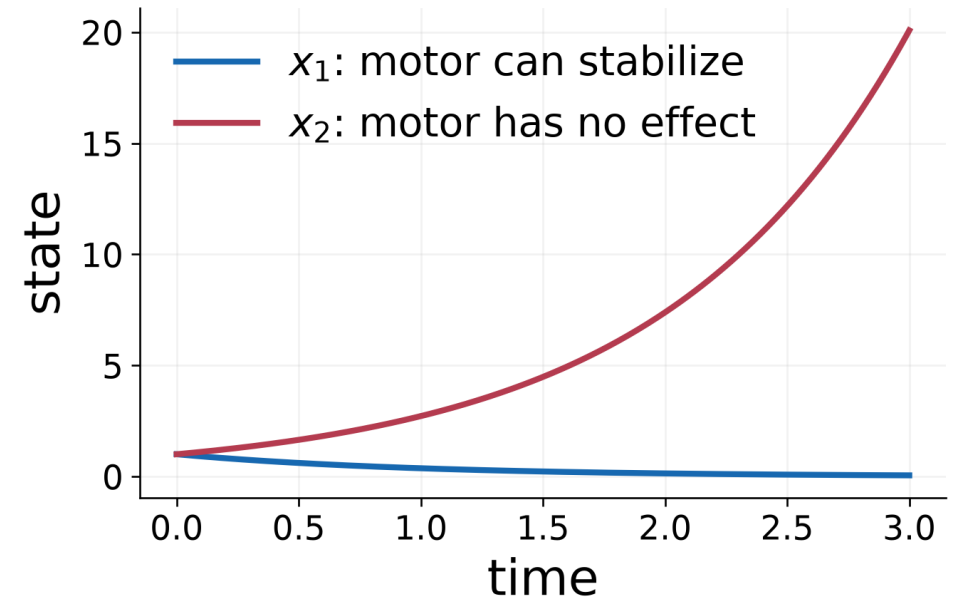
$$\dot{x}_1 = x_1 + u, \quad \dot{x}_2 = x_2.$$

The motor acts on x_1 ; it has **no effect on x_2** .

Choose $u = -2x_1$: then $\dot{x}_1 = -x_1$, but $\dot{x}_2 = x_2$ still grows.

Controllable: control inputs can move the full state between arbitrary states in finite time. Arbitrary eigenvalue placement requires controllability.

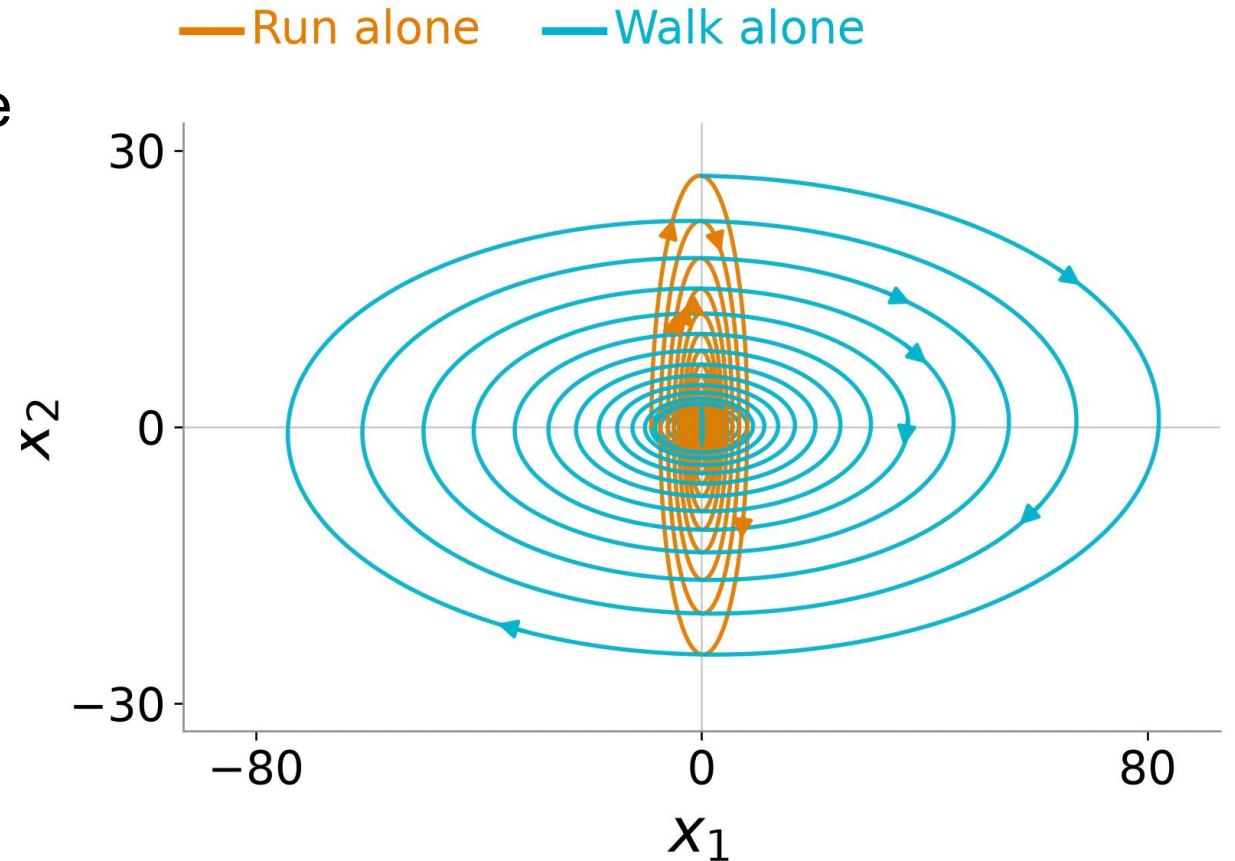
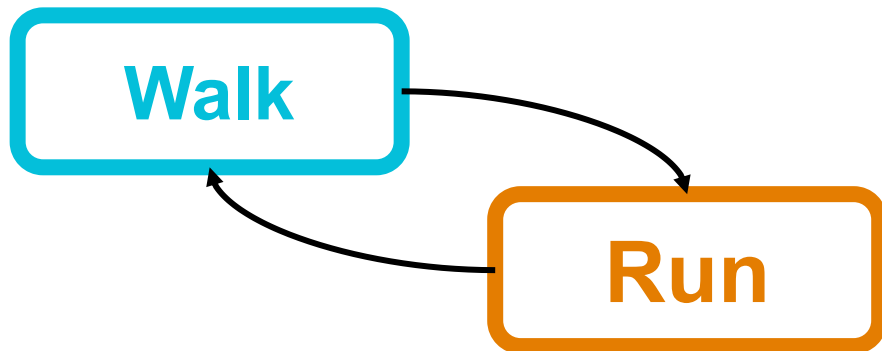
Stabilizable: every λ with $\operatorname{Re}\lambda \geq 0$ in the uncontrolled system can be influenced to be < 0 ; enough for asymptotic stabilization.



Final thoughts: can switching destabilize stable modes?

Suppose the Walk and Run models are each **asymptotically stable**.

Can switching between them make the combined motion unstable?

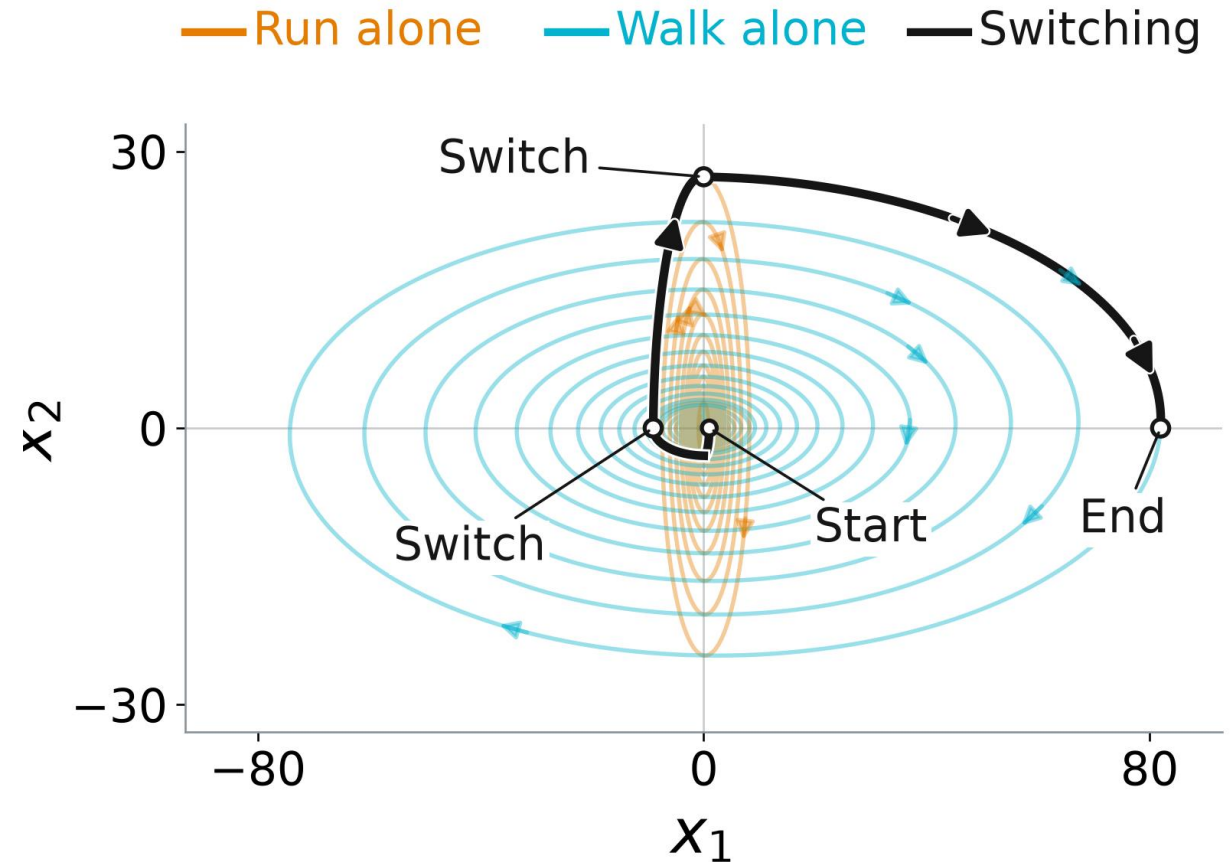
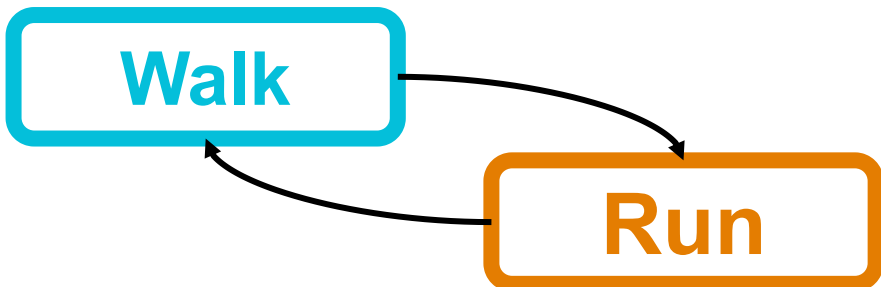


Yes: switching between stable modes can be unstable

A **hybrid model** combines continuous dynamics with **discrete modes** and transition rules.

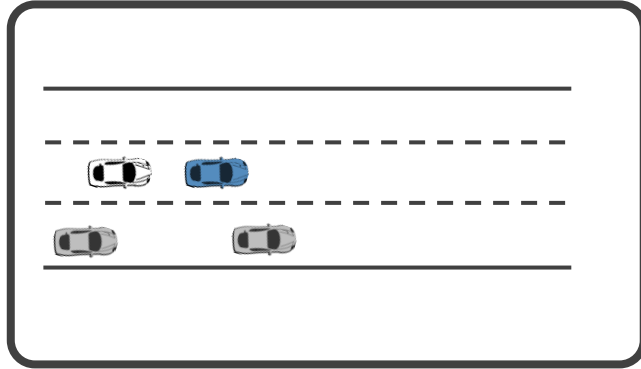
Each mode can shrink errors when run alone; switching can repeatedly select transient growth.

Hybrid stability also depends on the **switching rule**, not only the individual ODEs.



Same mode trajectories; the black path switches modes.

Hybrid dynamics



System dynamics

Physical plant

$$\dot{\mathbf{x}} = f(\mathbf{x}, u)$$

$$\mathbf{x}_{k+1} \approx \mathbf{x}_k + \Delta t f(\mathbf{x}_k, u_k)$$

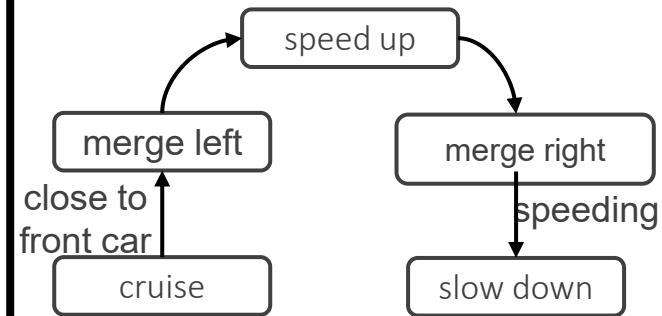
State

$$\mathbf{x} = [v_B, x_B, y_B, \delta_B, \theta_B]^T$$

Input

$$u = [a, r_\delta]^T, r_\delta = \dot{\delta}_B$$

Decision and control software



State: speed, position, steering, heading. Δt : sample time; k : sample index.

Summary

Lyapunov stability: every sufficiently small disturbance stays small.

Asymptotic stability: it also returns to the equilibrium.

Linear stability: all $\text{Re}\lambda < 0 \rightarrow$ asymptotically stable.

Zero real parts require checking coupling (Jordan blocks).

Eigenvalue placement: choose feedback gains to make closed-loop eigenvalues have negative real parts.

Switching: stable individual modes can still produce unstable motion.